

Hibernate Level 1 Cache (First-Level Cache)

1. Level 1 Cache

Level 1 Cache is enabled by default in Hibernate.

- Every **Session** has its own Level 1 Cache.
- It is created automatically when a Session is created.

2. Entity Storage

When you fetch an entity from the database using methods like **get()**, **load()**, or **getReference()**, Hibernate stores that entity in the Level 1 Cache after it has been loaded.

Database → Hibernate Session → Level 1 Cache

3. Same Entity in Same Session

If you request the **same entity with the same primary key (ID)** again within the same Session, Hibernate does **not** execute another SQL query. Instead, it returns the entity directly from the Level 1 Cache.

4. New Session

If you open a **new Session**, a new Level 1 Cache is created. Fetching the same entity with the same primary key (ID) using the new Session executes the SQL query again.

5. Advantages

- Reduces unnecessary database queries.
- Decreases database load.
- Improves application response time.
- Ensures only one managed entity instance exists for a given primary key within a Session.

Important Point

If the same entity with the same primary key (ID) is requested again within the same Session, Hibernate retrieves it from the Level 1 Cache instead of executing another SQL query.

Very Important Point

If the same entity, same primary key, and same Session already returned **null**, Hibernate will hit the database again.

Reason:

- Level 1 Cache stores only managed entity objects.
- If **session.get(Employee.class, 101)** returns **null**, it means no entity was found.
- Since there is no entity object to store, Level 1 Cache does **not** cache **null** as an entity.